ATLANTECH

INTELLIGENT PORT OPERATIONS

Screens Specification

Every page of the ATLANTECH port system: who uses it, what it contains, what users can do on it and how it behaves.

43	12	36	9
Pages	Roles	Web pages	Tablet pages

SCREENS · 1

Shared web pages

Used by most web roles. The content here is the base; each role adds its own items.

P-16 Port Login (Web)

Web · All web roles

P1

Purpose. Sign-in for office staff.

CONTENT

- Email and password fields
- Port name

ACTIONS

- Sign in

SPECIAL STATES

- Wrong credentials
- Account disabled
- Forced password change on first sign-in
- Session expired (cookie session)

Goes to P-17 or the role start page

P-17 Home Dashboard

Web · All web roles

P1

Purpose. Role start page summarising what needs the user's attention.

CONTENT

- Pending-work cards per role (see the widget table)
- Latest notifications
- Live changes
- Shortcuts to main pages

ACTIONS

- Open the related page from a card

SPECIAL STATES

- No pending work (empty)

Goes to Role pages

P-18 Unified Port State

Web · R2, R3, R4, R5, R7, R8, R10, R11

P2

Purpose. Live operational picture of the whole port.

CONTENT

- Berths and their status (map or list)
- Vessels present and expected
- Tugs and pilots: availability and occupation
- Running plans
- Filters: terminal, status, resource type

ACTIONS

- Open a plan or a resource

SPECIAL STATES

- Live update through SSE
- Terminal-scoped users see only their terminal

Goes to P-39, P-40

P-19 Notifications

Web + Tablet · All roles

P2

Purpose. User notifications: tasks, plans, reminders, deviations.

CONTENT

- List: type, related entity, sent time, status (Sent / Failed / Escalated, Acknowledged as a property)
- Filters: status, type
- Types: task, plan, reminder, escalation, deviation, recommendation ready, delay re-plan, drift alert
- New-plan notice when a plan changes

ACTIONS

- Acknowledge
- Open related entity

SPECIAL STATES

- Failed and Escalated shown clearly; the system retries and escalates (no manual resend)
- Tablet: works offline from the cache; acknowledge goes through the Outbox

Goes to P-55, P-40, P-36

P-20 Access Denied

Web + Tablet · All roles

P1

Purpose. Shown when the user lacks permission for a page or terminal.

CONTENT

- Reason: permission or terminal scope
- Back to home button

ACTIONS

- Back to home

SPECIAL STATES

- Shown on 403 or an out-of-scope request

Goes to P-17 / P-54

P-21 Environmental Conditions

Web · All web roles

P2

Purpose. Weather, waves, tide and data freshness.

CONTENT

- Current: wind, gust, visibility, wave height and period, current, sea level, surge
- Hourly forecast table
- Freshness: last pull time, source (Primary / Fallback / Manual), age

ACTIONS

- Refresh now (on-demand pull, proposed)

SPECIAL STATES

- Stale data warning (older than 48 hours or not covering the horizon)
- Provider failure

Goes to P-37

SCREENS · 2

Shared tablet pages

Used by the Marine Pilot, Tug Master and CT Supervisor. Offline-first.

P-52 Tablet Login

Tablet · R6, R9, R12

P1

Purpose. Sign-in on the device and re-authentication when the token expired offline.

CONTENT

- Username and password fields
- Port name
- Connection state

ACTIONS

- Sign in
- Change password (first sign-in)

SPECIAL STATES

- Token expired offline: sign in again, then send pending events of the same user only
- Pending events of another user are kept and not sent
- Wrong credentials
- Account disabled (403)
- Offline: sign-in impossible, clear message
- Sessions are per device

Goes to P-54

P-53 Port Identification

Tablet · R6, R9, R12

P1

Purpose. First-launch screen confirming the port the app connects to.

CONTENT

- Port name and id (read from the server)
- Minimum app version check

ACTIONS

- Continue

SPECIAL STATES

- Server unreachable: message and retry
- Wrong port (proposed)
- Offline: use last saved info
- Version below minimum: go to P-57

Goes to P-52

P-54 My Tasks

Tablet · R6, R9, R12

P1

Purpose. Executor home: assigned tasks (or terminal operations), from the cache and without internet.

CONTENT

- Connection and sync bar
- Task cards: vessel, berth, time, task type, status badge, sync indicator
- Live updates (WebSocket)
- Plan-changed notice
- Supervisor: terminal operations list

ACTIONS

- Open task
- Pull to refresh (proposed)

SPECIAL STATES

- Offline: works from the cache
- No tasks (empty)
- Loading and error

Goes to P-55, P-56, P-19

P-55 Task Details & Status Update

Tablet · R6, R9, R12

P1

Purpose. Show one task and record its actual progress.

CONTENT

- Task data: vessel, time, resources (berth, tugs, pilot)
- Acknowledge button (pressed by the executor)
- Large status buttons: In progress and Executed only
- System labels: Cancelled, Overridden, Updated
- "Saved on device" then "Sent" indicator
- New-plan notice

ACTIONS

- Acknowledge (through the Outbox)
- Update status

SPECIAL STATES

- Offline: saved on the device first
- Rejected event (for example, plan cancelled): reason shown, actions stop
- 409: current server state shown

Goes to P-54, P-56

P-56 Sync Status & Outbox

Tablet · R6, R9, R12

P1

Purpose. Pending, sent and rejected events.

CONTENT

- Connection state
- Event list: status (Pending / Sending / Done / Rejected), time recorded on device, rejection reason
- Counters
- Warning: Done events cleaned after 30 days; pending events lost if the device is wiped

ACTIONS

- Retry (proposed)

SPECIAL STATES

- No pending events
- Token expired: go to P-52 and keep events
- Only the signed-in user's events

Goes to P-52, P-54

P-57 App Update Required

Tablet · R6, R9, R12

P2

Purpose. Blocks use of an old app version.

CONTENT

- Message
- Current and required version

ACTIONS

- Update (store link proposed; production uses MDM)

SPECIAL STATES

- Triggered by 426 or the minimum version
- Sync and all actions blocked; Outbox kept
- No way back to app screens

Goes to -

P-58 Settings & Session

Tablet · R6, R9, R12

P1

Purpose. Account, device and app data; end the session.

CONTENT

- Account: name, email, role
- Port, app version, device data
- Language switch (Arabic / English with RTL)

ACTIONS

- Log out
- Open P-56

SPECIAL STATES

- Warn before logout when events are pending, then allow it
- Online logout revokes the token; offline logout deletes the local token only
- Pending events kept and marked with the owner

Goes to P-56, P-52

SCREENS · 3

Port Admin

Users, master data, settings, audit and AI governance.

P-23 User Management

Web · R2 P1

Purpose. List and manage port user accounts.

CONTENT

- Filters: role, terminal, status
- Table: name, email, role, terminal, active

ACTIONS

- Create user
- Deactivate
- Open user

SPECIAL STATES

- Deactivating yourself or the last Port Admin is refused (409)

Goes to P-24

P-24 User Details

Web · R2 P1

Purpose. Identity, role and terminal scope of one user.

CONTENT

- Name, email, status
- One active role (role history kept with an end date)
- Terminal scope (for terminal-scoped roles)
- Registered devices summary
- One-time password shown once after creation

ACTIONS

- Save
- Assign / end role
- Deactivate
- Open devices

SPECIAL STATES

- Field errors
- Terminal required for terminal-scoped roles
- Deactivating yourself or the last Port Admin refused (409)

Goes to P-23, P-25

P-25 Registered Devices

Web · R2 P1

Purpose. Devices linked to user sessions.

CONTENT

- Table: user, device type, app version, last seen, session state

ACTIONS

- Revoke one device
- Revoke all tokens of a user

SPECIAL STATES

- Confirmation dialog

Goes to P-24

P-26 Port Profile & Navigation Channel

Web · R2 P1

Purpose. Port data, boundary and channel waypoints.

CONTENT

- Port profile: name, boundary, time zone, coordinates
- Navigation channel: name, depth, width, chart datum
- Waypoints table in order: order, latitude, longitude
- Map editor for boundary and waypoints (proposed)

ACTIONS

- Save
- Add / reorder / remove waypoint

SPECIAL STATES

- Field errors

Goes to P-29

P-27 Terminal Management

Web · R2 P1

Purpose. Port terminals, their type and cargo types.

CONTENT

- Table: name, type (Container / Non-Container), cargo types, berth count, status
- Form: name, boundary, type, cargo types

ACTIONS

- Create
- Edit

SPECIAL STATES

- Non-Container terminals can be recorded; their operations pages are future work

Goes to P-28

P-28 Berth Management

Web · R2 P1

Purpose. Berth records and status.

CONTENT

- Table: code, terminal(s), length, draft depth, max LOA, status, location
- Form: same fields; one or more terminals (shared berths allowed)

ACTIONS

- Create
- Edit
- Deactivate

SPECIAL STATES

- Max LOA must fit within the berth length
- Warning: edits never change a confirmed plan; disabling a berth in a confirmed plan refused (409)

Goes to P-27, P-29

P-29 Port Audit & Change History

Web · R2 P1

Purpose. Who changed what and when in port and operational data.

CONTENT

- Filters: actor, entity type, action, date
- Table: time, actor, entity, action
- Details: before / after
- Refused attempts shown as Rejected attempt rows

ACTIONS

- Search and filter
- Open details

SPECIAL STATES

- No results

Goes to -

P-30 AI Model Governance

Web · R2

P2

Purpose. Evaluation and drift of the local AI (Part 3 MVP).

CONTENT

- Current model versions (champion per model) from the registry
- Evaluation metrics (C1) on trusted data only
- Drift measurements (C2) and drift alerts
- Note: all metrics are from labelled synthetic data until real operations start

ACTIONS

- Review (read-only, proposed)

SPECIAL STATES

- No measurements yet (empty)

Goes to -

P-62 Operational Settings

Web · R2

P2

Purpose. Port-level operational settings.

CONTENT

- Deviation thresholds per operation type
- Reminder lead time (default 2 hours)
- Typed settings editor (port_setting)
- Port shifts tab

ACTIONS

- Save

SPECIAL STATES

- Default
- Validation errors
- Saved

Goes to -

SCREENS · 4

Port Authority Staff

Vessel journey: P-31 → P-33 → P-34 / P-35 → P-36 → P-37 → P-38 → P-40.

P-31 BAPLIE Files

Web · R3
P1

Purpose. Upload BAPLIE files and follow their processing.

CONTENT

- Upload area (drag and drop)
- Table: file name, vessel, request, uploaded by, time, parse status, file status (Active / Replaced)
- Diagnostics panel per file
- Failure reason in the row

ACTIONS

- Upload BAPLIE
- Correct staged data and re-validate
- Withdraw
- Re-upload
- Open vessel and cargo data

SPECIAL STATES

- Auto-match by IMO and ETA within 24 hours; 422 when 0 or many matches
- Duplicate file refused (409)
- Parsing in progress
- Parsed
- Quarantined with diagnostics
- Failed with reason
- Withdraw refused if a confirmed plan used the file (409)
- Manifest hidden from other terminals

Goes to P-33, P-35

P-33 Vessel & Cargo Data

Web · R3
P1

Purpose. Vessel data and the container list extracted from the BAPLIE.

CONTENT

- Vessel: IMO, name, LOA, beam, draft, type
- Container table: number, size and type, weight, hazard class, stowage (bay / row / tier), load port, discharge port
- Count summary (proposed)
- Search and filters

ACTIONS

- Edit vessel particulars
- Create berthing request from this data

SPECIAL STATES

- No containers
- Incomplete manifest lines flagged (they block customs approval)

Goes to P-31, P-35

P-34 Berthing Requests

Web · R3
P2

Purpose. List of berthing requests (created manually by staff).

CONTENT

- Filters: status (state-diagram statuses), terminal, vessel, ETA
- Table: request number, vessel, requested terminal, status badge, created by, created at
- Live status updates

ACTIONS

- New request
- Open request

SPECIAL STATES

- No requests

Goes to P-35, P-36

P-35 Berthing Request Form

Web · R3
P1

Purpose. Create or edit a berthing request.

CONTENT
- Vessel by IMO (check digit) or from BAPLIE data
- Per-call particulars (drafts, ETA, Cb)
- Cargo lines
- Requested terminal
- Vessel and cargo summary (proposed)

ACTIONS
- Save
- Resubmit (after Needs revision)
- Cancel request (while allowed)

SPECIAL STATES
- Field errors
- Editing locked after the allowed step

Goes to P-36

P-36 Berthing Request Details

Web · R3
P1

Purpose. Full record and decision point for a request across its life cycle.

CONTENT
- Header: vessel and status badge
- Timeline: from, to, by, time
- Vessel and cargo summary
- Preliminary approval panel
- Customs approval panel
- Part 1 run status (Queued / Running / Failed)
- Links to entry window, recommendation and plan (by status)

ACTIONS
- Preliminary approve / Return for revision / Reject
- Record customs approval
- Reopen entry window (with reason)
- Edit / Cancel (by status)

SPECIAL STATES
- Actions shown by status
- Customs approval blocked while a manifest line is incomplete
- Part 1 failed: recompute

Goes to P-37, P-38, P-40

P-37 Entry-Time Window Review

Web · R3
P2

Purpose. Review proposed entry windows after customs approval (human gate 1, AI Part 1).

CONTENT
- Environmental summary and data freshness
- Timeline: safe vs blocked periods with the limiting factor (wind, visibility, waves, tide, UKC)
- List of proposed windows
- Peak congestion indicator
- Constraints are hard: blocked periods cannot be edited

ACTIONS
- Confirm window (then the AI generates the plan)
- Reject all windows
- Recompute
- Enter manual environmental data

SPECIAL STATES
- Outcome: No feasible window
- Outcome: Environmental data unavailable
- Part 1 running / failed after 3 attempts
- Stale data (older than 48 hours) with warning
- Confirm refused if not the latest proposal or no longer passing (409)

Goes to P-38, P-21

P-38 Recommendation Review

Web · R3

P1

Purpose. Review port-wide recommendations (berth, tugs, pilot) and turn one into a confirmed plan (human gate 2, AI Part 2).

CONTENT

- Ranked options: berth, time, tugs, pilot, objective score
- Service-time estimate (Phase 1 formula)
- Delay risk: "Not assessed" until a rule is approved
- Rationale text and rationale factors; binding constraints
- Source badge (AI / FIFO)
- Version history (Pending review, Modified by staff, Replaced)
- Override panel per component with mandatory reason
- Regeneration banner when resource state changes

ACTIONS

- Select option
- Override (with reason)
- Regenerate (before confirmation)
- Confirm plan

SPECIAL STATES

- FIFO fallback
- No feasible plan (request stays Entry window confirmed; Regenerate)
- Regenerating
- Generation failed
- Override rejected by constraint (422)
- Confirm failed: availability re-check (409, rolled back)
- Regenerate after confirmation refused (409)
- Review and confirmation by Port Authority Staff only

Goes to P-40

P-39 Operational Plans

Web · R3, R2; R10 and R11 read-only

P1

Purpose. List of confirmed and running plans.

CONTENT

- Filters: vessel, status, date
- Table: plan, vessel, status (Confirmed / Completed / Cancelled), date
- Live updates

ACTIONS

- Open plan

SPECIAL STATES

- CT Planner and CT Administrator: read-only, own terminal only

Goes to P-40

P-40 Operational Plan Details

Web · R3, R2; R10 and R11 read-only

P1

Purpose. One plan with its components, assignments and actual progress.

CONTENT

- Plan and vessel summary
- Components: berth, tugs, pilot, terminal, each with assigned resource, status, last update and notification status
- Actual progress timeline
- Deviation indicator and link
- Version history
- Delay re-plan proposal panel

ACTIONS

- Cancel plan (with reason; choose whether the call continues; releases resources)
- Safety exception (Port Admin only, mandatory reason, after a start)
- Confirm or discard a delay re-plan proposal
- Open deviation

SPECIAL STATES

- Cancel confirmation dialog
- Normal cancel refused after execution started
- Re-planned (new version)
- CT Planner and CT Administrator: view only

Goes to P-38, P-34, P-42

P-41 Deviation Records

Web · R3
P1

Purpose. Deviations between planned and actual.

CONTENT

- Filters: vessel, terminal, resource type, date
- Table: plan, component, planned, actual, detection time, reason, trust (Normal / Clock-skew suspect)

ACTIONS

- Filter
- Open deviation

SPECIAL STATES

- No deviations

Goes to P-42

P-42 Deviation Details

Web · R3
P1

Purpose. One deviation: planned vs actual and its reason.

CONTENT

- Planned vs actual side by side
- Reason (nullable)
- Related plan
- Trust flag

ACTIONS

- Open plan

SPECIAL STATES

- Empty reason

Goes to P-40

SCREENS · 5

Pilotage and tug departments

Resources and assignments.

P-43 Pilot Roster & Availability

Web · R4; R5 read-only

P1

Purpose. Pilots, ranks and shift schedule.

CONTENT

- Table: name, rank, shift start and end, status
- Rank limits (vessel LOA per rank)
- Availability calendar (proposed)
- Pilot details drawer (proposed)

ACTIONS

- Add pilot
- Remove
- Promote (change rank)
- Edit availability
- Edit rank limits

SPECIAL STATES

- Officer: read-only
- Warning: edits never change a confirmed plan; disabling a pilot in a confirmed plan refused (409)

Goes to P-44, P-18

P-44 Pilotage Assignments

Web · R5

P1

Purpose. Pilot components of confirmed plans waiting for assignment or assigned.

CONTENT

- List: vessel, berth, time window, proposed pilot, assigned pilot, status
- Pilot picker with availability, rank vs LOA and shift checks

ACTIONS

- Confirm proposed pilot
- Replace pilot

SPECIAL STATES

- Reminders T+10 and T+20, escalation T+30
- Availability or rank conflict (422 / 409)
- No pending tasks

Goes to P-43

P-45 Tugboat Fleet

Web · R7; R8 read-only

P1

Purpose. The tugboat fleet.

CONTENT

- Table: name, power, capacity, status, assigned tug master
- Add and edit form

ACTIONS

- Add
- Edit
- Remove

SPECIAL STATES

- Officer: read-only
- Warning: edits never change a confirmed plan; disabling a tug in a confirmed plan refused (409)

Goes to P-46

P-46 Tug Master Management

Web · R7; R8 read-only

P1

Purpose. Tug master accounts, shifts and tug links.

CONTENT

- Table: name, email, linked tug or unlinked, shift schedule
- Add account form

ACTIONS

- Add
- Remove
- Link / unlink tug (one tug per tug master)

SPECIAL STATES

- Officer: read-only
- Double link refused

Goes to P-45

P-47 Tug Assignments

Web · R8
P1

Purpose. Towage components of confirmed plans waiting for assignment or assigned.

CONTENT

- List: vessel, berth, time window, proposed tug master and tugs, assigned, status
- Picker with availability and shift checks

ACTIONS

- Confirm proposal
- Replace

SPECIAL STATES

- Reminders T+10 and T+20, escalation T+30
- Conflict (422 / 409)
- No pending tasks

Goes to P-45, P-46

SCREENS · 6

Container terminal

Terminal-scoped pages for the CT Administrator and CT Planner.

P-48 CT Equipment & Cranes

Web · R10
P1

Purpose. Terminal equipment and productivity.

CONTENT

- Table: category (equipment role), quantity, available quantity, status
- Specifications per type (STS, RTG, Reach stacker, Top-lift, Forklift, Truck)
- Cargo-type compatibility
- Productivity profiles tab (no overlapping periods)

ACTIONS

- Add
- Edit
- Remove

SPECIAL STATES

- Scoped to the user's terminal

Goes to P-49

P-49 Container Yard Structure

Web · R10
P1

Purpose. Yard structure used by placement recommendations.

CONTENT

- Hierarchy: yard → zones → blocks → slots
- Zone: type (General / Reefer / DG / Empty / OOG), code, capacity, occupancy
- Reefer: available and total power points, power type
- DG: allowed IMO classes
- OOG: max length, width, height, weight
- Slot: bay, row, tier, code, status

ACTIONS

- Add / edit / remove zone, block, slot
- Bulk slot creation (proposed)

SPECIAL STATES

- Many slots: grouped view and search

Goes to P-48

P-50 Yard Placement Recommendations

Web · R11
P2

Purpose. Container placement recommendations for vessels served at the terminal.

CONTENT

- Filters: status, vessel
- Table: vessel, generated time, status (Requested / Generating / Generation failed / Generated / Accepted / Rejected / Exported to TOS)

ACTIONS

- Open recommendation

SPECIAL STATES

- Scoped to the user's terminal

Goes to P-51

P-51 Yard Placement Review

Web · R11
P2

Purpose. Review one yard recommendation and record the result.

CONTENT

- Proposed zones / slots per container group
- Rationale and confidence (heuristic method)
- Capacity impact on zones (proposed)
- TOS-ready view

ACTIONS

- Accept
- Adjust (each adjustment re-validated)
- Reject
- Record exported to TOS

SPECIAL STATES

- Adjustment refused by a yard rule (422)
- Generation failed with reason

Goes to P-50

SCREENS · 7

Marine safety

Constraints applied before recommendations.

P-59 Safety Constraints

Web · R13

P1

Purpose. Safety and feasibility constraints applied before recommendations.

CONTENT

- Table: terminal, constraint type, value, unit, valid from, valid to
- Environmental limits (wind, waves, visibility), chart datum, minimum UKC
- Filter: active / upcoming / expired (proposed)
- Tide constituent set activation with datum check

ACTIONS

- Create
- Edit
- Activate tide constituent set

SPECIAL STATES

- Constraints are hard
- Overlapping constraints refused
- Datum mismatch refused

Goes to -